ASSIGNMENT-2

Take one Domain and draw the graph (Normal distribution) (Empirical rule)

1.Introduction to the Domain and Normal Distribution

1.1 Introduction

In statistics and data science, many real-world variables follow a special type of distribution called the Normal Distribution. It is also known as the Bell Curve because of its symmetrical bell-shaped graph.

For this report, we select the domain:

Heights of Students in a College

Student heights usually follow a normal distribution because:

- Most students have average height
- Very few students are extremely short
- Very few students are extremely tall

This natural balance creates a symmetrical bell-shaped curve.

1.2 What is Normal Distribution?

A Normal Distribution is a continuous probability distribution that:

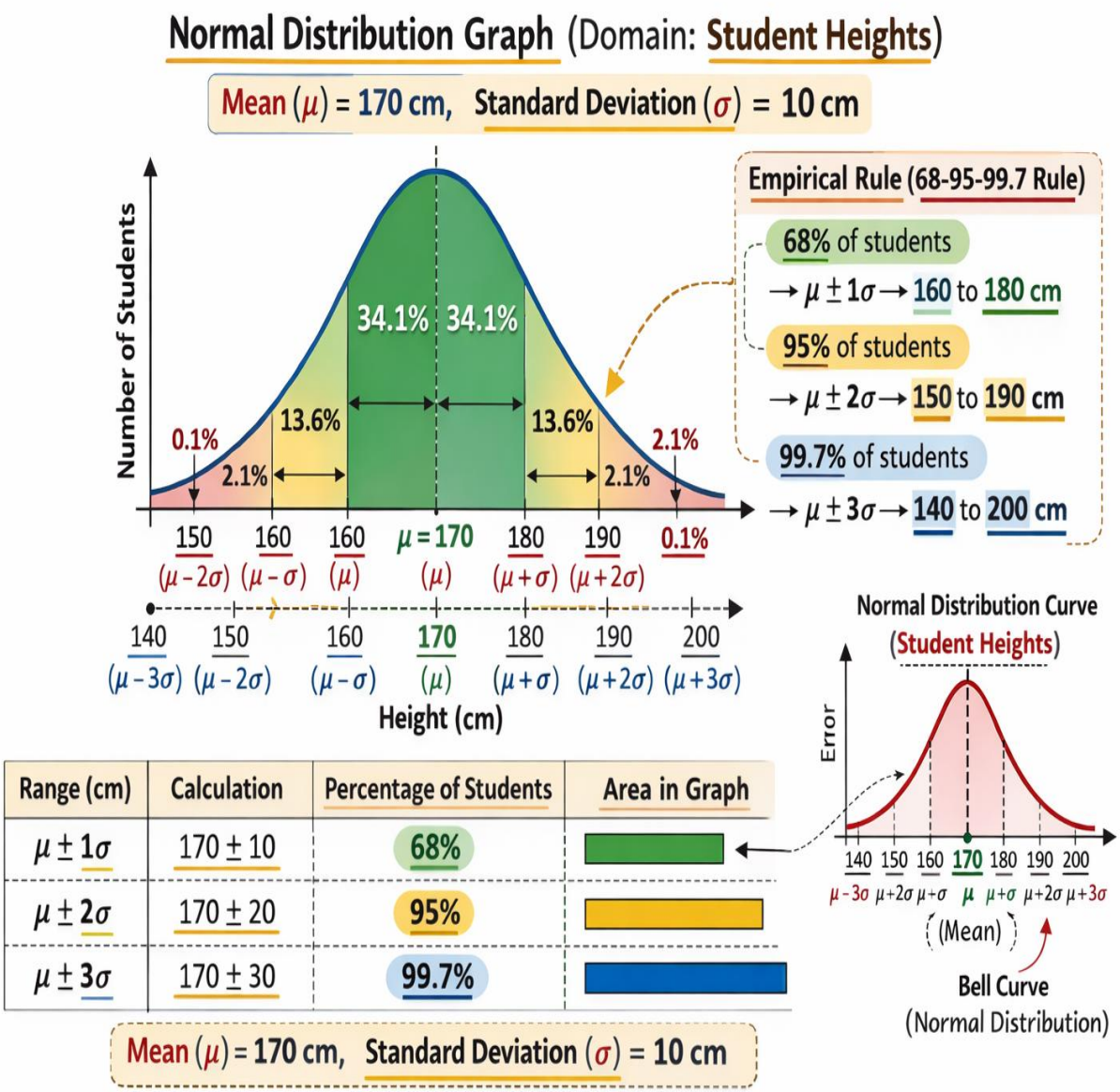
- Is symmetric around the mean (μ)
- Has a bell-shaped curve
- Mean = Median = Mode
- Defined by two parameters:
  - Mean (μ)
  - Standard Deviation (σ)

1.3 Formula of Normal Distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Where:

- μ = Mean
- σ = Standard deviation
- x = Data value



2.Given Data for Student Heights

Let us assume:

- Mean height (μ) = 170 cm
- Standard deviation (σ) = 10 cm
- Total students = 100

2.1 Understanding Mean (μ)

The mean is the average height of students.

$$\mu = 170 \text{ cm}$$

This means most students are around 170 cm tall.

2.2 Understanding Standard Deviation (σ)

Standard deviation tells us how spread out the heights are.

$$\sigma = 10 \text{ cm}$$

- Small σ → heights close to mean
- Large σ → heights widely spread

2.3 Shape of the Bell Curve

In the graph:

- Center point = 170 cm
- Curve is symmetric
- Left side = shorter students
- Right side = taller students

The highest point of the curve is at the mean.

3.Empirical Rule (68–95–99.7 Rule)

The Empirical Rule applies only to normal distribution.

It states:

- **68%** of data lies within **1 standard deviation**
- **95%** of data lies within **2 standard deviations**
- **99.7%** of data lies within **3 standard deviations**

3.1 First Standard Deviation (μ ± 1σ)

$$170 \pm 10 = 160 \text{ to } 180$$

About **68 students out of 100** have height between 160 cm and 180 cm.

3.2 Second Standard Deviation (μ ± 2σ)

$$170 \pm 20 = 150 \text{ to } 190$$

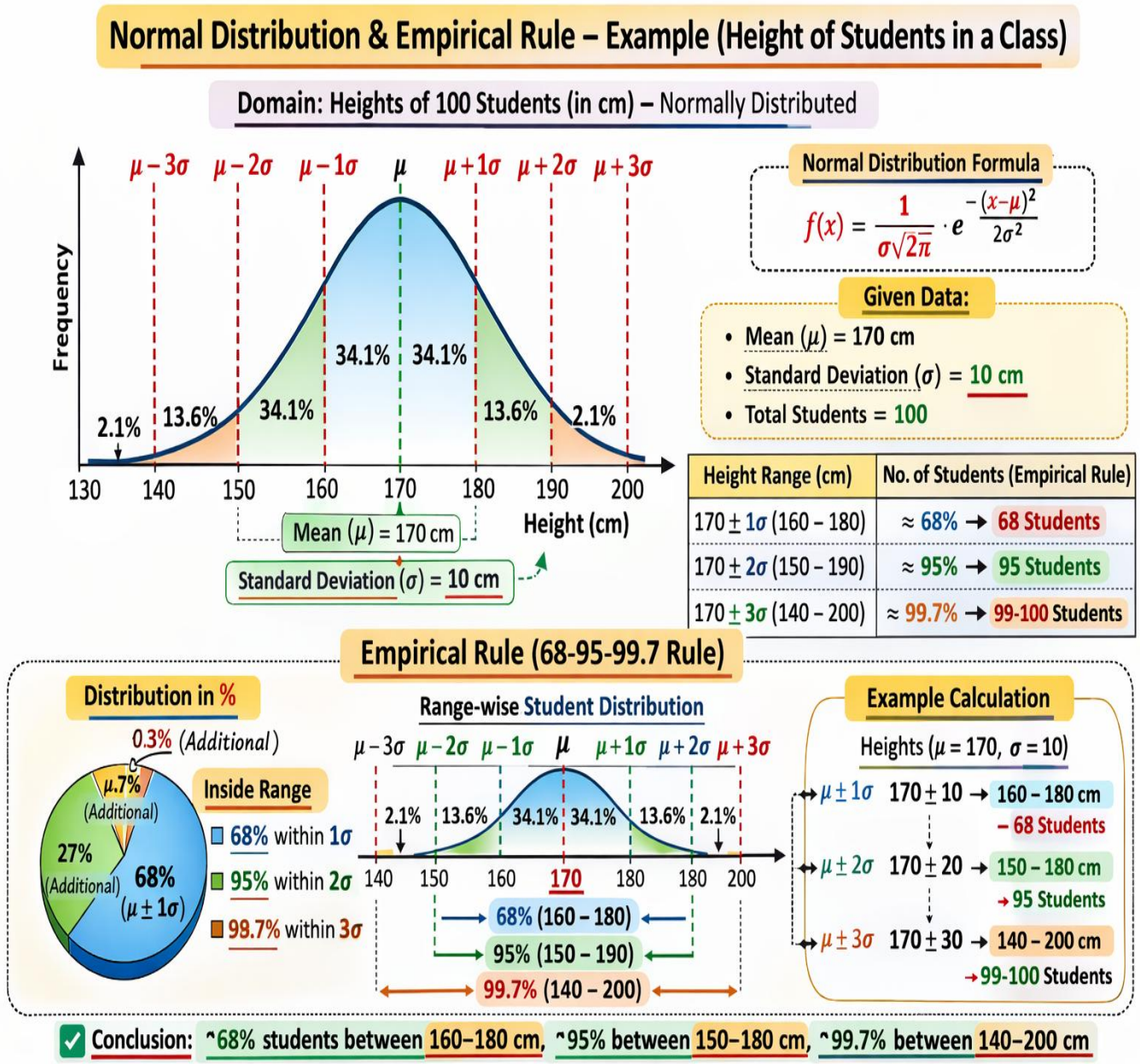
About **95 students out of 100** have height between 150 cm and 190 cm.

3.3 Third Standard Deviation (μ ± 3σ)

$$170 \pm 30 = 140 \text{ to } 200$$

About **99 or 100 students** have height between 140 cm and 200 cm.

Very few students fall outside this range.



4.Graphical Interpretation

4.1 Bell Curve Sections

The normal distribution graph is divided into sections:

- Between μ and $\mu + 1\sigma \rightarrow 34.1\%$
- Between μ and $\mu - 1\sigma \rightarrow 34.1\%$
- Between 1σ and $2\sigma \rightarrow 13.6\%$
- Between 2σ and $3\sigma \rightarrow 2.1\%$
- Beyond $3\sigma \rightarrow 0.1\%$

4.2 Why is it Symmetrical?

Because:

- Distribution depends on squared term $(x - \mu)^2$
- Positive and negative distances treated equally

Thus:

- Short and tall students are equally distributed around mean.

4.3 Importance of the Empirical Rule

The empirical rule helps us:

- Estimate probabilities quickly
- Understand spread of data
- Detect outliers
- Make statistical predictions

5. Applications and Conclusion

5.1 Applications in Real Life

Normal distribution is used in:

- Student heights
- Exam scores
- IQ scores

- Blood pressure
- Machine learning models
- Error analysis
- Quality control

5.2 In Machine Learning

Normal distribution helps in:

- Data preprocessing
- Feature scaling
- Gaussian Naïve Bayes
- Statistical modeling
- Hypothesis testing

5.3 Outliers

Students below 140 cm or above 200 cm are rare.

These are called:

Outliers

They represent less than 0.3% of data.

5.4 Final Conclusion

The heights of students follow a **Normal Distribution** with:

- Mean (μ) = 170 cm
- Standard Deviation (σ) = 10 cm

Using the **Empirical Rule**:

- 68% students → 160–180 cm
- 95% students → 150–190 cm
- 99.7% students → 140–200 cm

Thus, the normal distribution provides a powerful way to understand real-world data and make accurate predictions